

Experiment 02:

7. Laboratory Exercise

1. Attach screenshots of: WHOIS, theHarvester, DNSenum, Dmitry and WhatWeb command output. Describe your observations under each SS in detail. Use indicators such as highlight, colour, and box for this purpose.

1. whois --help

Output:

```
(student@kali)-[~]
$ whois --help
Usage: whois [OPTION]... OBJECT...

-h HOST, --host HOST    connect to server HOST
-p PORT, --port PORT    connect to PORT
-I                      query whois.iana.org and follow its referral
-H                      hide legal disclaimers
    --verbose           explain what is being done
    --no-recursion      disable recursion from registry to registrar servers
    --help              display this help and exit
    --version           output version information and exit

These flags are supported by whois.ripe.net and some RIPE-like servers:
-l                      find the one level less specific match
-L                      find all levels less specific matches
-m                      find all one level more specific matches
-M                      find all levels of more specific matches
-c                      find the smallest match containing a mnt-irt attribute
-x                      exact match
-b                      return brief IP address ranges with abuse contact
-B                      turn off object filtering (show email addresses)
-G                      turn off grouping of associated objects
-d                      return DNS reverse delegation objects too
-i ATTR[,ATTR]...       do an inverse look-up for specified ATTRIBUTES
-T TYPE[,TYPE]...       only look for objects of TYPE
-K                      only primary keys are returned
-r                      turn off recursive look-ups for contact information
-R                      force to show local copy of the domain object even
                        if it contains referral
-a                      also search all the mirrored databases
-s SOURCE[,SOURCE]...   search the database mirrored from SOURCE
-g SOURCE:FIRST-LAST    find updates from SOURCE from serial FIRST to LAST
-t TYPE                request template for object of TYPE
-v TYPE                request verbose template for object of TYPE
-q [version|sources|types] query specified server info
```

Observation:

The command displayed the help menu for whois and listed the available command options. It showed different flags for choosing servers, managing recursion, and setting query options.

2. whois cloudflare.com

Output:

```
(student@kali)-[~/Shivam]
$ whois cloudflare.com
Domain Name: CLOUDFLARE.COM
Registry Domain ID: 1564775704_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.cloudflare.com
Registrar URL: http://www.cloudflare.com
Updated Date: 2023-03-19T11:04:43Z
Creation Date: 2009-08-06T17:08:34Z
Registry Expiry Date: 2032-08-06T17:08:34Z
Registrar: Cloudflare, Inc.
Registrar IANA ID: 1910
Registrar Abuse Contact Email: registrar-abuse@cloudflare.com
Registrar Abuse Contact Phone: +1.6503198930
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: FISH.NS.CLOUDFLARE.COM
Name Server: WELLS.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-24T05:55:09Z <<<
```

Observation:

The command displayed the domain registration details for **cloudflare.com**, including the creation date, expiry date, registrar information, active name servers, and domain status codes.

3. theHarvester -h

Output:

```
$ theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*                               *
* [ _ _ _ _ _ ] [ _ _ _ _ _ ] *
* [ _ _ _ _ _ ] [ _ _ _ _ _ ] *
* [ _ _ _ _ _ ] [ _ _ _ _ _ ] *
* [ _ _ _ _ _ ] [ _ _ _ _ _ ] *
*                               *
* theHarvester 4.10.1          *
* Coded by Christian Martorella *
* Edge-Security Research       *
* cmartorella@edge-security.com *
*                               *
*****
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-S START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLV]]

theHarvester is used to gather open source intelligence (OSINT) on a company or domain.

options:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN    Company name or domain to search.
  -l, --limit LIMIT      Limit the number of search results, default=500.
  -S, --start START      Start with result number X, default=0.
  -p, --proxies          Use proxies for requests, enter proxies in proxies.yaml.
  -s, --shodan          Use Shodan to query discovered hosts.
  --screenshot SCREENSHOT
                        Take screenshots of resolved domains specify output directory: --screenshot output_directory
  -e, --dns-server DNS_SERVER
                        DNS server to use for lookup.
  -t, --take-over        Check for takeovers.
  -r, --dns-resolve [DNS_RESOLVE]
                        Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.
  -n, --dns-lookup       Enable DNS server lookup, default False.
  -c, --dns-brute        Perform a DNS brute force on the domain.
  -f, --filename FILENAME
                        Save the results to an XML and JSON file.
  -w, --wordlist WORDLIST
                        Specify a wordlist for API endpoint scanning.
  -a, --api-scan         Scan for API endpoints.
  -q, --quiet            Suppress missing API key warnings and reading the api-keys file.
  -b, --source SOURCE    baidu, bevigil, bitbucket, brave, bufferoverrun, builtwith, censys, certspotter, chaos, commoncrawl, crim
```

Observation:

The command displayed the help menu for **theHarvester**. It showed the available command options, supported search sources, and different settings for information gathering.

4. theHarvester -d cloudflare.com -b urlscan

Output:

```
(student@kali)-[~/Shivam]
$ theHarvester -d cloudfare.com -b urlscan
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [ASCII Art]
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: cloudfare.com

[*] Searching Urlscan.

[*] ASNS found: 11
-----
AS13335
AS14618
AS15169
AS16276
AS16509
AS19318
AS201814
AS209242
AS271354
AS3215
AS33182
```

```
[*] Interesting Urls found: 4
_____
http://a.nel.cloudflare.com/
https://challenges.cloudflare.com/
https://cloudflare.com/
https://cloudflare.com/cdn-cgi/trace

[*] IPs found: 32
_____
104.16.123.96
104.16.124.96
104.16.132.229
107.190.129.90
```

Observation:

The command searched **Urlscan** for information related to **cloudflare.com**. It found **11 ASNs**, **4 interesting URLs**, and **32 IP addresses** linked to the target domain.

5. dnsenum -h

Output:

```
(student@kali)-[~/Shivam]
$ dnsenum -h
dnsenum VERSION:1.3.1
Usage: dnsenum [Options] <domain>
[Options]:
Note: If no -f tag supplied will default to /usr/share/dnsenum/dns.txt or
the dns.txt file in the same directory as dnsenum
GENERAL OPTIONS:
  --dnsserver <server>      Use this DNS server for A, NS and MX queries.
  --enum                    Shortcut option equivalent to --threads 5 -s 15 -w.
  -h, --help                Print this help message.
  --noreverse               Skip the reverse lookup operations.
  --nocolor                 Disable ANSIColor output.
  --private                 Show and save private ips at the end of the file domain_ips.tx
  --subfile <file>          Write all valid subdomains to this file.
  -t, --timeout <value>    The tcp and udp timeout values in seconds (default: 10s).
  --threads <value>        The number of threads that will perform different queries.
  -v, --verbose             Be verbose: show all the progress and all the error messages.
GOOGLE SCRAPING OPTIONS:
  -p, --pages <value>      The number of google search pages to process when scraping nam
the default is 5 pages, the -s switch must be specified.
  -s, --scrap <value>      The maximum number of subdomains that will be scraped from Goo
BRUTE FORCE OPTIONS:
  -f, --file <file>        Read subdomains from this file to perform brute force. (Takes
  -u, --update <alg|rlz>   Update the file specified with the -f switch with valid subdom
      a (all)               Update using all results.
      g                     Update using only google scraping results.
      r                     Update using only reverse lookup results.
      z                     Update using only zonetransfer results.
  -r, --recursion           Recursion on subdomains, brute force all discovered subdomains
WHOIS NETRANGE OPTIONS:
  -d, --delay <value>      The maximum value of seconds to wait between whois queries, th
  -w, --whois               Perform the whois queries on c class network ranges.
                          **Warning**: this can generate very large netrangles and it wi
REVERSE LOOKUP OPTIONS:
  -e, --exclude <regex>    Exclude PTR records that match the regexp expression from reve
OUTPUT OPTIONS:
  -o --output <file>       Output in XML format. Can be imported in MagicTree (www.gremwe
```

Observation:

The command displayed the help menu for **dnsenum**. It showed the available command options for selecting DNS servers, setting thread count, and saving the scan results in XML format.

6. dnsenum cloudflare.com

Output:

```
(student@kali)~[~/Shivam]
$ dnsenum cloudflare.com
dnsenum VERSION:1.3.1

cloudflare.com

Host's addresses:

cloudflare.com.      210      IN      A       172.67.211.231
cloudflare.com.      210      IN      A       104.21.77.216

Wildcard detection using: pjuhxhfrfbvbs

pjuhxhfrfbvbs.cloudflare.com.  300      IN      CNAME   cloudflare.com.
cloudflare.com.                111      IN      A       104.16.133.229
cloudflare.com.                111      IN      A       104.16.132.229

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

Wildcards detected, all subdomains will point to the same IP address
Omitting results containing 104.16.133.229, 104.16.132.229.
Maybe you are using OpenDNS servers.

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

```
Name Servers:

fish.ns.cloudflare.com.      86263    IN      A       108.162.194.252
fish.ns.cloudflare.com.      86263    IN      A       162.159.38.252
fish.ns.cloudflare.com.      86263    IN      A       172.64.34.252
wells.ns.cloudflare.com.     86263    IN      A       108.162.195.125
wells.ns.cloudflare.com.     86263    IN      A       162.159.44.125
wells.ns.cloudflare.com.     86263    IN      A       172.64.35.125

Mail (MX) Servers:

mailstream-east.mxrecord.io.  29       IN      A       172.65.213.128
mailstream-west.mxrecord.io.  29       IN      A       172.65.220.210
mailstream-central.mxrecord.mx.  29      IN      A       172.65.64.78
mailstream-canary.mxrecord.io.  210      IN      A       172.65.237.73

Trying Zone Transfers and getting Bind Versions:

Trying Zone Transfer for cloudflare.com on fish.ns.cloudflare.com ...
AXFR record query failed: FORMERR

Trying Zone Transfer for cloudflare.com on wells.ns.cloudflare.com ...
AXFR record query failed: FORMERR

Brute forcing with /usr/share/dnsenum/dns.txt:

*.cloudflare.com.      210      IN      CNAME   cloudflare.com.
1003.cloudflare.com.   210      IN      CNAME   cloudflare.com.
1025.cloudflare.com.   210      IN      CNAME   cloudflare.com.
1027.cloudflare.com.   210      IN      CNAME   cloudflare.com.
1029.cloudflare.com.   210      IN      CNAME   cloudflare.com.
1037.cloudflare.com.   210      IN      CNAME   cloudflare.com.
```

cloudflare.com class C netranges:

```
104.21.77.0/24
172.67.211.0/24
```

Performing reverse lookup on 512 ip addresses:

```
0 results out of 512 IP addresses.
```

cloudflare.com ip blocks:

```
done.
```

Observation:

The command scanned **cloudflare.com** and displayed its IP addresses, name servers, and mail servers. It also checked for zone transfers, searched for subdomains, and listed the Class C network ranges.

7. dmitry -h

Output:

```
(student@kali)-[~/Shivam]
$ dmitry -h
Deepmagic Information Gathering Tool
"There be some deep magic going on"

dmitry: invalid option -- 'h'
Usage: dmitry [-winsepfb] [-t 0-9] [-o %host.txt] host
  -o      Save output to %host.txt or to file specified by -o file
  -i      Perform a whois lookup on the IP address of a host
  -w      Perform a whois lookup on the domain name of a host
  -n      Retrieve Netcraft.com information on a host
  -s      Perform a search for possible subdomains
  -e      Perform a search for possible email addresses
  -p      Perform a TCP port scan on a host
* -f      Perform a TCP port scan on a host showing output reporting filtered ports
* -b      Read in the banner received from the scanned port
* -t 0-9  Set the TTL in seconds when scanning a TCP port ( Default 2 )
*Requires the -p flagged to be passed
```

Observation:

The command displayed an invalid option message and then showed the help menu for **dmitry**. It listed the available command options for WHOIS lookups, subdomain searches, Netcraft queries, and port scanning.

8. dmitry -h

Output:

```
(student@kali)-[~/Shivam]
$ dmitry -h
Deepmagic Information Gathering Tool
"There be some deep magic going on"

dmitry: invalid option -- 'h'
Usage: dmitry [-winsepfb] [-t 0-9] [-o %host.txt] host
  -o      Save output to %host.txt or to file specified by -o file
  -i      Perform a whois lookup on the IP address of a host
  -w      Perform a whois lookup on the domain name of a host
  -n      Retrieve Netcraft.com information on a host
  -s      Perform a search for possible subdomains
  -e      Perform a search for possible email addresses
  -p      Perform a TCP port scan on a host
  * -f      Perform a TCP port scan on a host showing output reporting filtered ports
  * -b      Read in the banner received from the scanned port
  * -t 0-9 Set the TTL in seconds when scanning a TCP port ( Default 2 )
  *Requires the -p flagged to be passed
```

Observation:

The command displayed an invalid option message and then showed the help menu for **dmitry**. It listed the available command options for WHOIS lookups, subdomain searches, Netcraft queries, and port scanning.

9. dmitry cloudflare.com -s

Output:

```
(student@kali)-[~/Shivam]
$ dmitry cloudflare.com -s
Deepmagic Information Gathering Tool
"There be some deep magic going on"

HostIP:104.21.77.216
HostName:cloudflare.com

Gathered Subdomain information for cloudflare.com
-----
Searching Google.com:80 ...
Searching Altavista.com:80 ...
Found 0 possible subdomain(s) for host cloudflare.com, Searched 0 pages containing 0 results

All scans completed, exiting
```

Observation:

The command searched for subdomain information related to **cloudflare.com**. It did not find any subdomains, and the scan completed successfully.

10. dmitry cloudflare.com -wise

Output:

```
(student@kali)-[~/Shivam]
$ dmitry cloudflare.com -wise
Deepmagic Information Gathering Tool
"There be some deep magic going on"

HostIP:172.67.211.231
HostName:cloudflare.com

Gathered Inet-whois information for 172.67.211.231


---



inetnum:          171.76.0.0 - 172.80.127.255
netname:          NON-RIPE-NCC-MANAGED-ADDRESS-BLOCK
descr:            IPv4 address block not managed by the RIPE NCC
remarks:          

---


remarks:
remarks:          For registration information,
remarks:          you can consult the following sources:
remarks:
remarks:          IANA
remarks:          http://www.iana.org/assignments/ipv4-address-space
remarks:          http://www.iana.org/assignments/iana-ipv4-special-registry
remarks:          http://www.iana.org/assignments/ipv4-recovered-address-space
remarks:
remarks:          AFRINIC (Africa)
remarks:          http://www.afrinic.net/ whois.afrinic.net
remarks:
remarks:          APNIC (Asia Pacific)
remarks:          http://www.apnic.net/ whois.apnic.net
remarks:
remarks:          ARIN (Northern America)
remarks:          http://www.arin.net/ whois.arin.net
remarks:
remarks:          LACNIC (Latin America and the Carribean)
remarks:          http://www.lacnic.net/ whois.lacnic.net
remarks:          

---


country:          EU # Country is really world wide
admin-c:          IANA1-RIPE
```

Observation:

The command gathered **Inet-WHOIS** information for the IP address of **cloudflare.com**. It displayed the IP address range, network details, and the regional registry sources for the IP.

11. whatweb cloudflare.com

Output:

```
(student@kali)-[~/Shivam]
$ whatweb cloudflare.com
http://cloudflare.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[cloudflare], I
mmonHeaders[report-to,nel,cf-ray,alt-svc], Via-Proxy[HTTP/1.1 forward.http.proxy:3128]
https://cloudflare.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[cloudflare],
ommonHeaders[report-to,nel,cf-ray,alt-svc]
https://www.cloudflare.com [200 OK] Cookies[__cf_bm,_ga,cfz_adobe,cfz_google-analytics_v4,kndctr_8/
HTTPServer[cloudflare], HttpOnly[__cf_bm,cfz_adobe,cfz_google-analytics_v4], IP[104.16.124.96], M
son,module,text/javascript,true], Strict-Transport-Security[max-age=31536000; includeSubDomains],
icy,cross-origin-opener-policy,cross-origin-resource-policy,permissions-policy,referrer-policy,x-c
-XSS-Protection[1; mode=block]
```

Observation:

The command scanned **cloudflare.com** and identified the web server details. It showed the website redirects, IP address, cookies, security headers, and other technologies used by the website.

2. Record observations of: Domain details, DNS records, Website technologies identified.

Domain Details	
Domain Name:	GITHUB.COM
Registry Domain ID:	1264983250_DOMAIN_COM-VRSN
Registrar:	MarkMonitor Inc.
Creation Date:	2007-10-09T18:20:50Z
Updated Date:	2024-09-07T09:16:32Z
Registry Expiry Date:	2026-10-09T18:20:50Z

DNS Record	
Domain Name:	github.com
IP Address:	20.207.73.82
Host Address:	github.com
Name Servers:	ns-1707.awsdns-21.co.uk, dns4.p08.nsone.net, dns2.p08.nsone.net, dns3.p08.nsone.net, ns-421.awsdns-52.com, dns1.p08.nsone.net, ns-520.awsdns-01.net, ns-1283.awsdns-32.org
Mail Servers:	github-com.mail.protection.outlook.com (52.101.40.6, 52.101.11.2, 52.101.50.13, 52.101.194.0)

Technologies Used	
Domain Name:	github.com
List of Technologies Used:	
HTTP Server:	GitHub
Web Protocol:	HTTP/2
Content Language:	en-US
Website Features:	HTML5, Redirect Location
Open Graph:	Open-Graph-Protocol
Search & Scripting:	OpenSearch, JavaScript
Security Headers:	X-XSS-Protection
Cookies:	_gh_sess, _octo, logged_in
Network Information:	Country – United States, IP Address – 20.207.73.82
Website Title:	<i>GitHub · Change is constant. GitHub keeps you ahead.</i>